

Prime-Square Closure of the Frozen Determinant-Two Logarithmic Shadow

Full Möbius Signs, Fixed Periodic Masks, and the Growing-Data Firewall

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Abstract

Let

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = 2$$

be fixed primitive positive affine forms, and let ρ be a fixed bounded periodic sequence. We prove the unconditional fixed-data theorem

$$\sum_{x/\omega(x) < z \leq x} \frac{\mu(D(z))\mu(V(z))\rho(z)}{z} = o(\log \omega(x))$$

for every $1 \leq \omega(x) \leq x$ with $\omega(x) \rightarrow \infty$. The analytic input is Tao's proved logarithmically averaged two-point theorem for fixed affine forms. To close the two squarefree masks without changing its quantifiers, we use the Boolean prime-square cutoff

$$S_P(m) = \prod_{p \leq P} (1 - \mathbf{1}_{p^2|m}).$$

For fixed P it has a finite signed CRT expansion, whereas its full-squarefree error has logarithmic relative mass $O(P^{-1}) + o(1)$. The order of limits is essential: P is fixed before $x \rightarrow \infty$, and only then tends to infinity. The result is a genuine fixed-shift arithmetic shadow, but it is logarithmically weighted and frozen. It proves no uniformity in growing coefficients, physical weights or phases, no ordinary all-prefix power saving, and no positive L2 input.

1 Fixed determinant-two data

Fix integers $a, s \geq 1$ and $d, u \in \mathbb{Z}$ satisfying

$$(d, s) = (u, a) = (a, s) = 1, \quad su - ad = 2. \tag{1}$$

After a fixed translation of z , if necessary, assume

$$D(z) = d + sz > 0, \quad V(z) = u + az > 0$$

throughout the asymptotic range. The determinant condition makes the coefficient pairs (s, d) and (a, u) nonparallel. TPC-127 identifies these forms with the literal fixed-two Liouville carrier, while TPC-128 opens their squarefree indicators by a magnitude cutoff [2, 3].

Let $\rho : \mathbb{Z} \rightarrow \mathbb{C}$ be fixed, periodic of period M , and satisfy $|\rho| \leq 1$. The words “fixed” and “periodic” are theorem quantifiers: neither M nor the values of ρ may grow with x in the theorem below.

2 A Boolean prime-square cutoff

Let

$$P^\# = \prod_{p \leq P} p$$

and define

$$S_P(m) = \prod_{p \leq P} (1 - \mathbf{1}_{p^2|m}). \quad (2)$$

Lemma 2.1 (Exact finite expansion). *For every positive integer m ,*

$$S_P(m) = \sum_{k|P^\#} \mu(k) \mathbf{1}_{k^2|m} \in \{0, 1\}. \quad (3)$$

Consequently

$$S_P(D(z))S_P(V(z)) = \sum_{k, \ell | P^\#} \mu(k)\mu(\ell) \mathbf{1}_{k^2|D(z)} \mathbf{1}_{\ell^2|V(z)}. \quad (4)$$

Proof. Expand the finite Euler product in (2). Its value is one exactly when no prime $p \leq P$ has $p^2 \mid m$, and is zero otherwise. Multiplying the two one-variable expansions gives (4). $\square$

For a compatible pair (k, ℓ) , the two square divisibilities select one residue modulo $\text{lcm}(k, \ell)^2$. In the determinant-two primitive setting, compatibility in fact forces the conditions recorded in TPC-128, including $(k, \ell) = 1$, but the following upper bound does not require equality of moduli.

Proposition 2.2 (Correct combined modulus). *After also resolving the period M , every nonempty component of (4) is a union of residue classes modulo*

$$L_{M,k,\ell} = \text{lcm}(M, k^2, \ell^2),$$

and

$$L_{M,k,\ell} \leq M \text{lcm}(k, \ell)^2 \leq M(P^\#)^2. \quad (5)$$

There are at most $4^{\pi(P)}$ ordered component pairs.

Proof. Intersect the three periodic conditions. Since $\text{lcm}(k^2, \ell^2) = \text{lcm}(k, \ell)^2$, the first inequality follows from $\text{lcm}(M, L) \leq ML$. Both k and ℓ divide $P^\#$, proving the second inequality. There are $2^{\pi(P)}$ divisors of $P^\#$, independently in each coordinate. $\square$

Remark 2.3 (Why no fourth power appears). The two divisors k, ℓ are drawn from the same prime set. The universal bound is $M(P^\#)^2$, not $M(P^\#)^4$. Writing a product $k^2\ell^2$ is harmless only after compatibility has forced $(k, \ell) = 1$; even then $k\ell \mid P^\#$.

Lemma 2.4 (Residue and content preserve nonparallelity). *Let $L \geq 1$ and $0 \leq r < L$. On the progression $z = r + Lt$, put*

$$c_D = (sL, d + sr), \quad c_V = (aL, u + ar).$$

After extracting these positive contents, the two primitive forms

$$\tilde{D}(t) = \frac{sL}{c_D}t + \frac{d + sr}{c_D}, \quad \tilde{V}(t) = \frac{aL}{c_V}t + \frac{u + ar}{c_V}$$

have coefficient determinant

$$\det(\tilde{D}, \tilde{V}) = \frac{L(su - ad)}{c_D c_V} = \frac{2L}{c_D c_V} \neq 0. \quad (6)$$

Proof. The two displayed quotients are primitive by the definitions of c_D, c_V . Before content extraction their determinant is

$$(sL)(u + ar) - (aL)(d + sr) = L(su - ad) = 2L.$$

Dividing the two coefficient pairs by their contents proves (6). In particular $c_D c_V \mid 2L$, and no CRT residue can make the reduced forms parallel. $\square$

3 A tail suited to a two-limit theorem

Lemma 3.1 (One-form large-prime-square tail). *Let $L(z) = b + cz$ be fixed, primitive and positive on an integer interval I of length N , and suppose $L(z) \leq Y$. Then*

$$\sum_{z \in I} |\mu^2(L(z)) - S_P(L(z))| \leq \sum_{\substack{p > P \\ p \leq \sqrt{Y}}} \left(\frac{N}{p^2} + 1 \right) \ll_L \frac{N}{P} + \sqrt{Y}. \quad (7)$$

Proof. If $\mu^2(m) \neq S_P(m)$, then $p^2 \mid m$ for some prime $p > P$. For $p \nmid c$, the condition $p^2 \mid L(z)$ selects one residue modulo p^2 ; for $p \mid c$, primitivity makes it insoluble. Summing $N/p^2 + 1$ and using $\sum_{n > P} n^{-2} \ll P^{-1}$ proves the claim. $\square$

Unlike the magnitude-truncated divisor sum, S_P is Boolean. Therefore

$$\begin{aligned} & \left| \mu^2(D)\mu^2(V) - S_P(D)S_P(V) \right| \\ & \leq |\mu^2(D) - S_P(D)| + |\mu^2(V) - S_P(V)|. \end{aligned} \quad (8)$$

No divisor-function maximum is introduced.

Proposition 3.2 (Terminal logarithmic tail). *For the fixed forms above,*

$$\sum_{x/\omega < z \leq x} \frac{|\mu^2(D(z))\mu^2(V(z)) - S_P(D(z))S_P(V(z))|}{z} \ll_{D,V} \frac{\log \omega}{P} + 1 \quad (9)$$

uniformly for $1 < \omega \leq x$.

Proof. Partition the terminal interval into dyadic pieces $(T, 2T]$, with at most one truncated end piece. On one such piece, lemma 3.1 and equation (8) and $z^{-1} \leq T^{-1}$ give

$$\sum_{T < z \leq 2T} \frac{\text{tail}(z)}{z} \ll_{D,V} P^{-1} + T^{-1/2}.$$

There are $O(1 + \log \omega)$ pieces, while their $T^{-1/2}$ terms form a geometric sum when ordered from the smallest scale. The finitely many pieces below a fixed positivity threshold are absorbed in the constant. $\square$

4 Full-Möbius logarithmic closure

We use the following source theorem only with its native fixed-data quantifiers. For fixed positive affine forms

$$L_i(z) = a_i z + b_i, \quad a_1 b_2 - a_2 b_1 \neq 0,$$

Tao proved

$$\sum_{x/\omega(x) < z \leq x} \frac{\lambda(L_1(z))\lambda(L_2(z))}{z} = o(\log \omega(x)) \quad (10)$$

whenever $1 \leq \omega(x) \leq x$ and $\omega(x) \rightarrow \infty$ [1].

Theorem 4.1 (Fixed determinant-two full-squarefree closure). *Under (1), for every fixed bounded periodic ρ and every admissible ω ,*

$$\sum_{x/\omega(x) < z \leq x} \frac{\mu(D(z))\mu(V(z))\rho(z)}{z} = o(\log \omega(x)). \quad (11)$$

Proof. Fix P . Insert (4) and split ρ into its finitely many residue classes. On each nonempty combined residue write $z = r + Lt$. The two forms become fixed affine forms in t . Extract their fixed positive contents. Lemma 2.4 gives their exact reduced determinant $2L/(c_D c_V)$, so Tao’s nonparallelity hypothesis survives both the residue split and content extraction. Complete multiplicativity extracts fixed Liouville signs.

Discard the finitely many terms with $t \leq 0$. The transformed terminal interval is

$$\frac{x/\omega - r}{L} < t \leq \frac{x - r}{L}.$$

If its lower endpoint is at least one, take the ratio of the two endpoints; otherwise start at $t = 1$. In either case this gives an admissible transformed terminal ratio $\omega'(x) \rightarrow \infty$ with

$$\log \omega'(x) = \log \omega(x) + O_{r,L}(1).$$

Equation (10), together with

$$\frac{1}{r + Lt} = \frac{1}{L} \frac{1}{t} + O_{r,L}(t^{-2}),$$

then gives $o(\log \omega)$ for every component. The number of components is fixed because P, M, D, V are fixed.

Since $\mu = \mu^2 \lambda$, the difference between the full Möbius correlation and this truncated correlation is bounded by proposition 3.2. After division by $\log \omega$,

$$\limsup_{x \rightarrow \infty} \frac{|\text{full sum}|}{\log \omega(x)} \ll_{D,V} \frac{1}{P}.$$

Now let $P \rightarrow \infty$. □

Remark 4.2 (Order of limits). The theorem does not choose a prescribed $P = P(x)$. For each accuracy request, one first fixes a sufficiently large P , and then takes x sufficiently large depending on all fixed data. This is exactly why theorem 4.1 is not a growing-family theorem.

5 Claim boundary and next gate

Statement	Level and status
Prime-square identity and component bound	Exact finite theorem (L0).
Large-prime-square and logarithmic tails	Unconditional elementary theorem (L0).
Full $\mu(D)\mu(V)$, fixed periodic logarithmic cancellation	Unconditional frozen arithmetic shadow (L1).
Prescribed growing P , affine height, period, or CRT modulus	Not proved.
Physical weight, arbitrary additive phase, ordinary all-prefix power estimate	Not proved; actual L2 gate.
H3, 1/400, or a prime-pair theorem	Not proved.

TPC-137 closes a precise gap left by the finite-mask statement of TPC-129 [4]: the full squarefree masks can be returned for fixed data without claiming a quantitative cutoff rate. The next question is whether stronger quantitative results known for the consecutive shift-one carrier touch the active odd shift-two carrier. They do not do so by a parity-preserving relabeling.

References

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